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Notes & Tips

Assembly of highly diverse genes using degenerate oligonucleotides by temperature cascade

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ABSTRACT

The quality of a nucleotide-based library such as a synthetic antibody library is highly dependent on the diversity available. Diversity can be generated using degenerate oligonucleotides introduced during gene assembly. Conventional approaches to gene assembly are not efficient for oligonucleotides with long stretches of degeneracy. We propose an efficient alternative for simultaneous introduction of three randomized regions in a synthetic antibody gene via temperature cascading. The strategy takes advantage of DNA reannealing kinetics. The strategy can be adopted for generating diversity of gene inserts during the construction of nucleotide-based libraries.

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The immune system applies a set of complex processes to generate a vastly diverse collection of antibodies against foreign proteins. To mimic the antibody generation processes in vitro, the construction of highly diverse synthetic antibody libraries involves the introduction of diversity in the complementarity-determining regions (CDRs) to a fixed framework [1,2]. The construction of a full antibody domain bottom up is commonly done using gene assembly methods [3]. Here, chemically synthesized oligonucleotides with degenerate trinucleotides were designed and assembled into a full-length gene sequence. These degenerate oligonucleotides replicated the required diversity synonymous with the diversity occurring in vivo. Variation in the length of degeneracy allowed further diversification of the antibodies [4]. This is the common approach for the generation of any highly diverse synthetic nucleotide-based library. The conventional one-pot gene assembly method allows a single-step mixture of all relevant oligonucleotides to anneal and assemble in a final PCR step [5] (Fig. 1a). Such methods are efficient for fixed sequence oligonucleotides but not necessary efficient for oligonucleotides with high degeneracy [6]. The inefficiency of assembly is a result of the higher complexity created from highly degenerate oligonucleotides. To assemble oligonucleotides with high degeneracy, some had proposed methods involving ligation of fragments to assemble a gene, which is time consuming and costly because of the requirement of phosphorylated oligonucleotides and also T4 DNA ligase [7]. However, we propose a new approach whereby a temperature cascade was introduced to improve the efficiency of assembly with degenerate oligonucleotides at fixed positions throughout the gene with low background.

In this study, we used the protocol to assemble a collection of synthetic antibody genes with high diversity for library cloning. The germ-line sequence that we used to design the oligonucleotides is VH3-23 (DP47), which was taken from VBASE2 and IMGT [8] (Supplementary Table 1). The full-length sequence was divided into seven segments, which were Framework 1, Framework 2, Framework 3, Framework 4 (JH4*02), CDR1, CDR2, and CDR3. Each of the CDRs consists of degenerate trinucleotides (MNN) with two complementary overlaps of 18 bp at the 5′ and 3′ ends [9]. The length of the MNNs in the CDRs is based on the most frequently occurring length of CDR and the conical confirmation for each antibody subfamily [10,11]. Restriction sites are introduced at the 5′ and 3′ ends for cloning. A pair of outermost primers was designed to amplify the assembled full-length sequence.

Our proposed PCR assembly approach starts with the temperature cascade method as shown in Fig. 1a. An equimolar amount of each segment (100 nM) was added to a total reaction of 20 μl, containing 20 mM Tris–HCl (pH 8.8), 10 mM $(NH_4)_2SO_4$, 10 mM KCl, 0.1 mg/ml bovine serum albumin (BSA), 0.1% (v/v) Triton X-100, 2 mM $MgSO_4$, 0.5 mM dNTP, and 1.25 U *Pfu* DNA polymerase (Fermentas, Vilnius, Lithuania). The mixture was heated at 95 °C for 2 min 30 s, then at 90 °C for 2 min 30 s, and then slowly cooled down to 85, 70, 60, 50, 40, 30, and 25 °C, for 10 min at each temperature. The temperature ramp was set at 0.1 °C/s, starting from 95 °C (Fig. 1a). This temperature cycling was carried out using a PTC-200 Thermo Cycler (MJ Research, Watertown, MA, USA).

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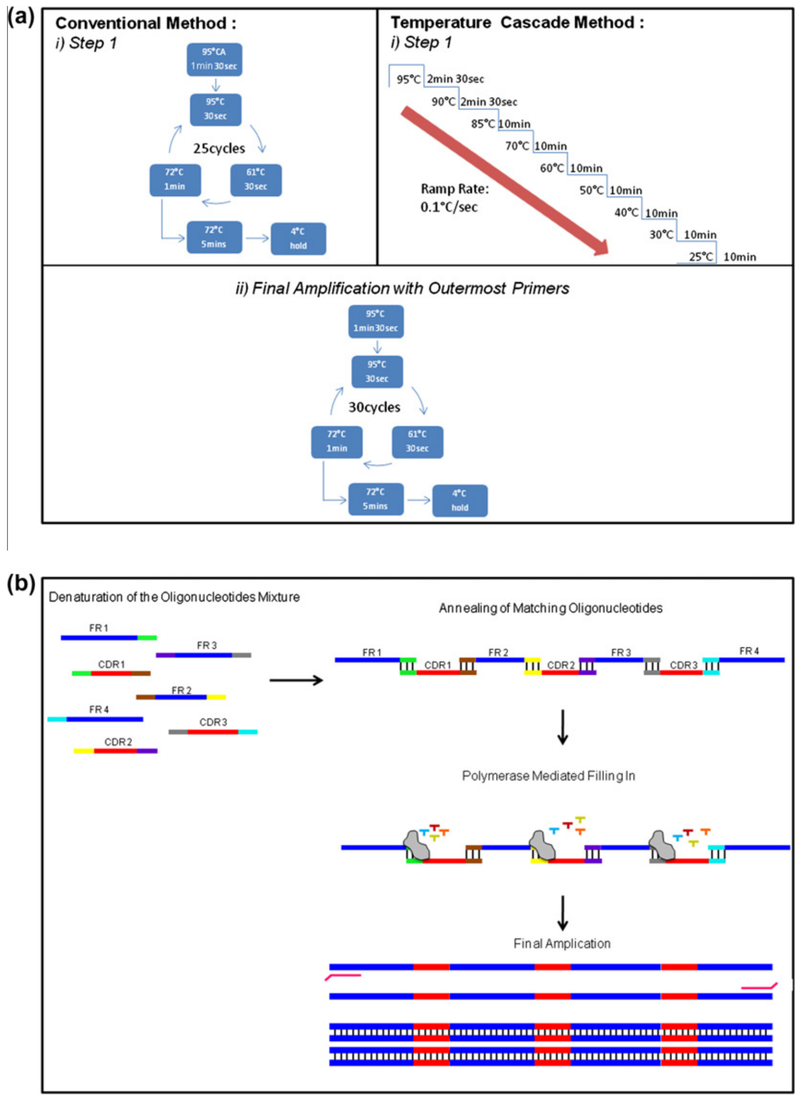

Fig.1. (a) Schematic representation of the proposed protocol, the temperature cascade method, and the conventional polymerase cycling assembly method. (i) First step of assembly. (ii) Final amplification of the full-length gene using outermost primers. (b) Illustration of the assembly of the designed oligonucleotides. At high temperature, the oligonucleotides' mobility is at its highest, which causes the oligonucleotides to be far apart from one another. As the temperature decreases at a slow ramp, the oligonucleotides will be less mobile and edge closer to one another. At this point, each of the oligonucleotides will find its complementary strand and hybridize at a higher accuracy. Once hybridized, the DNA polymerase starts to fill in the gap. Last, at the second step of assembly, final amplification of the full-length gene is carried out with a pair of outermost primers.

Temperature cycling was followed by amplification of the assembled full-length gene. Six microliters from the first reaction was mixed into 20 μl PCR mix and amplified with 10-fold concentration of the outermost primers. The amplification profile used was 95 °C for 1 min 30 s; 30 cycles of 95 °C for 30 s, 61 °C for 30 s, and 72 °C for 1 min; and last, 72 °C for 5 min (Fig. 1a). After the amplification, the assembled VH gene was subjected to gel electrophoresis (1.2% (w/v) agarose gel) and purified using a gel extraction kit according to the manufacturer's instructions (Qiagen, Hilden, Germany).

We had also tried to assemble our high-degeneracy oligonucleotides using the conventional method that was adapted from Stemmer's method [12] (Fig. 1a). A faint band with high background that was caused by unassembled or nonspecific assembly of oligonucleotides was obtained (Fig. 2b). Unlike the assembly of degenerate oligonucleotides, the assembly of fixed oligonucleotides of a single gene using Stemmer's method showed a very good band at the expected size (Fig. 2a). This showed the inefficiency of the method at assembling highly degenerate oligonucleotides. However, a strong band at the expected size (384 bp) was obtained with the temperature cascade (Fig. 2d). The slow-ramp temperature cascade allows the degenerate oligonucleotides to have sufficient time to find their complementary oligonucleotide and fully hybridize (Fig. 1b). The designed temperature cascade consists of many different temperatures. This set of temperatures allows the various oligonucleotides to anneal to their complementary strand at their own specific annealing temperature. The chances of obtaining high-GC-content sequences when using degenerate oligonucleotides is high. This can cause the formation of primer secondary structures such as hairpin loops and primer dimerization that could have a deleterious effect on the assembly [13]. Therefore, the mixture was first treated at high temperature to break the formation of any secondary structures. To further reduce the chances of mishybridization, the PCR mixture volume is recommended to be kept below 50 μl; we used a 20-μl total reaction mixture [14]. This protocol was designed to reduce negative influ-

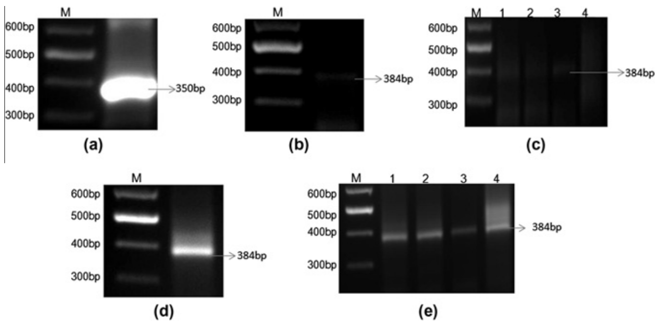

Fig.2. Agarose gel images. (a) Assembly of fixed and known gene using the conventional method; 100 bp plus ladder labeled as M. (b) Assembly of the high-degeneracy oligonucleotides using the conventional method; 100 bp plus ladder labeled as M. (c) Comparison of assembly without and with PCR enhancers using the temperature cascade method; 100 bp plus ladder labeled as M. Lane 1, sample without additive; lane 2, sample with BSA; lane 3, sample with DMSO; lane 4, sample with ET SSB. (d) Assembly of the high-degeneracy oligonucleotides using the temperature cascade method; 100 bp plus ladder labeled as M. (e) Comparison of assembly without and with PCR enhancers using the temperature cascade method; 100 bp plus ladder labeled as M. Lane 1, sample without additive; lane 2, sample with BSA; lane 3, sample with DMSO; lane 4, sample with ET SSB.

ences on the kinetics of the primer hybridization to achieve an improved binding of the degenerate oligonucleotides to the complementary strands [14].

Enhancers are commonly added to polymerase chain reactions to improve the yield and specificity. We also investigated the influence of adding PCR enhancers in the mixture. The PCR enhancers used were dimethyl sulfoxide (DMSO), BSA, and Extreme Thermophilic Single Strand Binding Protein, ET SSB (New England Biolabs, Ipswich, MA, USA), which were reported to be secondary structure destabilizing agents [15]. The use of these enhancers in PCR is to reduce the formation of homoduplex and heteroduplex DNA structures by destabilizing the DNA duplexes and enhance the stabilization of single-stranded DNA structures [16]. Subsequently, they also aid in reducing the background of the undesired assembled gene. However, we found that with the addition of the enhancers, no difference in the yield of the desired band was seen compared to the yield of sample without enhancers for both the conventional method and the proposed protocol (Fig. 2c and e). Thus, this temperature cascade method is able to generate a collection of diverse synthetic domain antibody genes with high degeneracy at fixed positions efficiently without the need of additives. Sequencing of the purified PCR products showed successful assembly of a diverse domain of antibody genes. The purified PCR products were digested with restriction enzymes *Nco*I and *Xho*I (New England Biolabs) and ligated to an in-house phagemid vector, pDAb. A total of 30 clones were sequenced (1st Base, Selangor Darul Ehsan, Malaysia) and the results indicated that close to 70% of the clones carry the full-length VH gene sequence with in-frame degenerate regions. However, because of the nature of the NNK degeneracy, wherein K represents G/T, there is the probability of a single amber stop codon (TAG) being present. A total of 11 clones were successfully cloned without amber stops (Supplementary Table 2), with another 9 clones showing the presence of the amber stop codon in the degenerate region.

In summary, the use of highly degenerate nucleotides in conventional PCR assembly is less preferable as it may disrupt the correct formation of full-length gene sequences. Thus, the presented temperature cascade protocol is highly efficient and enables easier generation of highly diverse synthetic nucleotide-based libraries other than antibodies.

Acknowledgments

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.ab.2012.08.025.

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